Alexa Warranty Vault

Voice-First Home Appliance Companion & Grounded FastMCP Server

Amazon Developer Hackathon
Submission Dossier & Specification

`PRIMARY TRACK: ALEXA+` `MINI CHALLENGE: OPEN SOURCE`

`LICENSE: MIT OPEN SOURCE`



GitHub Username: aaryansingh200617-cyber

License: MIT Open Source License (included in root)

Repository URL: https://github.com/aaryansingh200617-cyber/Alexa-Warranty-Vault

Contribution Branch: main (committed & pushed during hackathon)

1. Executive Summary & Inspiration

Every household manages dozens of appliances and electronics—from dishwashers and refrigerators to smart TVs, laptops, and coffee makers. When a device malfunctions, homeowners face a frustrating ordeal: paper receipts are lost, digital invoices are buried in email archives, warranty deadlines pass unnoticed, and filing a manufacturer claim requires digging for serial numbers and writing tedious formal emails.

Inspired by the natural, ambient voice intelligence of **Amazon Alexa** ("*Your Home, Your Love*"), **Alexa Warranty Vault** was engineered as an intelligent, voice-first home appliance companion. It proactively tracks warranty expirations, monitors coverage health with visual countdowns, and autonomously drafts airtight, professional warranty claim emails the moment an appliance exhibits issues.

> 🎯 **Primary Track Alignment: Alexa+**
>
> *"Build a self-hosted MCP server (spec 2025-11-25 or later, Streamable HTTP) or an Agent Skill. MCP is the open standard that powers Alexa+ integrations. Build a simulated Alexa+ experience in a web app using your preferred agentic tool."*

Alexa Warranty Vault directly satisfies and demonstrates the Alexa+ track requirements through two core pillars:

A. Self-Hosted Model Context Protocol (FastMCP) Server

- **Standard Compliance:** Implements a dedicated FastMCP server over **Streamable HTTP** (Server-Sent Events / SSE) at endpoint `/mcp` compliant with the modern MCP specification.
- **Deterministic Anti-Hallucination Grounding:** Alexa+ agents cannot guess purchase dates or fabricate coverage terms. Every warranty duration, status badge, and email draft is computed strictly through executable MCP tools connected to an ACID-compliant SQLite datastore.
- **Published MCP Tool Catalog:**

Tool Name	Input Parameters	Description & Execution Logic
`log_purchase`	`item_name`, `category`, `price`, `purchase_date`, `warranty_months`, `retailer`, `serial_number`	Registers appliance purchases, stores receipt metadata, and automatically calculates future expiration dates.
`check_warranty`	`query` (e.g., "dishwasher", "Sony TV")	Queries active coverage, computes days remaining until expiry, and returns status: *active*, *expiring_soon*, or *expired*.
`expiring_soon`	`days_ahead` (default: 30)	Scans household inventory for warranties nearing their expiration window to trigger proactive ambient alerts.
`draft_claim`	`product_name`, `issue_description`	Autonomously synthesizes a formal manufacturer claim email pre-filled with order ID, purchase date, serial number, and symptom analysis.

B. Simulated Alexa+ Web Experience

- **Voice-First Alexa Interaction:** Integrates the browser Web Speech API for voice recognition and fine-tuned natural Alexa speech synthesis, delivering hands-free assistance.
- **Hybrid Dual-Engine Architecture:**
  - *Online AI Mode:* Connects an LLM agent (Groq / OpenRouter) to the FastMCP server via tool-calling loops.
  - *Offline Edge Mode:* Built-in deterministic regex and date-engine for 100% private, zero-latency local operation without requiring any API keys or internet connection.
- **Segmented Sliding Toggle:** Seamlessly switches between Online AI and Offline Mode with real-time health telemetry (`/api/health`).

> **🌟 Mini Challenge Alignment: Open Source**
>
> *"Ship a new, additional open-source project (include an open-source license) or a contribution during the hackathon window. Include contribution URL, repo URL, GitHub username, and a short description of what you did, how it works, and why it matters."*

GitHub Username	`aaryansingh200617-cyber`
Repository URL	https://github.com/aaryansingh200617-cyber/Alexa-Warranty-Vault
Contribution Branch	https://github.com/aaryansingh200617-cyber/Alexa-Warranty-Vault/tree/main
Open-Source License	**MIT License** (Full permissive open-source license committed in root `LICENSE`)

What We Did

During the Amazon Developer Hackathon window, we developed and publicly released **Alexa Warranty Vault** as an open-source reference project. We authored the complete codebase from scratch, including the FastMCP server, the dual-router FastAPI agent bridge, the responsive React glassmorphic web frontend, Docker configuration, and comprehensive documentation. All development was pushed to GitHub with verified commit authorship by `aaryansingh200617-cyber` under the permissive MIT License.

How It Works

The open-source repository implements a clean, decoupled three-tier architecture:

1. **Frontend Layer (`web-app/`):** A React + Vite SPA styled with Tailwind CSS, utilizing Web Speech recognition and synthesis. Deployed with automated CI/CD on Vercel.
2. **Agent Orchestrator (`agent-api/`):** A FastAPI service providing dynamic CORS, session history memory, and an intelligent router that routes requests either to an LLM tool-calling loop or to a local rule engine.
3. **FastMCP Server (`mcp-server/`):** A standalone FastMCP server over Streamable HTTP with SQLite persistence, exposing standardized MCP tool definitions that any MCP-compliant client or Alexa+ runtime can connect to.

Why It Matters

As voice assistants evolve into agentic systems with the launch of Alexa+, developers need open-source, reproducible blueprints that illustrate how the Model Context Protocol (MCP) bridges generative AI with deterministic external tools. Alexa Warranty Vault demonstrates how to eliminate AI hallucinations in mission-critical applications (such as warranty dates and financial receipts) while ensuring consumer privacy through a reliable offline-first fallback. It serves as an accessible learning template and foundation for the global Alexa developer community.

2. System Architecture & Data Flow

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+-------------------------------------------------------------------------------+ | ALEXA+ SIMULATED WEB EXPERIENCE | | React (Vite) + Tailwind CSS + Web Speech API (Voice / Audio) | | Segmented Mode Toggle: [ ONLINE AI | OFFLINE ] | +-------------------------------------------------------------------------------+ | REST API / JSON (Over CORS / HTTPS) v +-------------------------------------------------------------------------------+ | AGENT ORCHESTRATION LAYER (FastAPI) | | - Session Memory (Multi-turn conversations) | | - Mode Router: | | [Online Mode] --> LLM Tool-Calling Loop (Groq / OpenRouter / OpenAI) | | [Offline Mode] --> Deterministic Regex & Date Inference Rule Agent | +-------------------------------------------------------------------------------+ | Streamable HTTP (FastMCP Client / SSE) v +-------------------------------------------------------------------------------+ | SELF-HOSTED FastMCP SERVER (Port 8765) | | Tools: log_purchase() | check_warranty() | expiring_soon() | draft_claim() | +-------------------------------------------------------------------------------+ | SQLite Database (Purchases, Warranties, Claims)
```

3. Key Features & Innovations

- 🎙️ **Ambient Voice Interaction:** Hands-free voice recognition and female Alexa voice synthesis tuned to match Amazon Echo's conversational persona.
- 🛡️ **Universal 1-Click Claim Drafting:** Works for any registered appliance (dishwashers, TVs, blenders, laptops). Extracts serial numbers, order IDs, and generates a structured, professional claim email ready for manufacturer submission.
- ⚡ **100% Offline Edge Capability:** Even during internet outages or without an LLM API key, homeowners can slide to Offline mode for zero-latency, private warranty tracking.
- ☁️ **Cloud-Native Deployment:** Fully containerized with Docker, backend hosted on Render, and frontend deployed on Vercel with automated GitHub CI/CD webhooks.

4. Quick Verification & Installation

Clone & Run Locally:

```
# 1. Clone repository git clone https://github.com/aaryansingh200617-cyber/Alexa-Warranty-Vault.git cd Alexa-Warranty-Vault # 2. Run with automated setup script start.bat # (Windows: starts MCP Server, Agent API, and Vite web UI) ./start.sh # (macOS/Linux)
```